Wealth & Health: 60 Years of Global Progress (1960-2023)

A Data Visualization Project

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A Personal Note and Reflection:

Before going into the details, let me say that the project really made me think. We all know, of course, that life expectancy increases, but seeing those dots move across the screen from year to year, watching as countries struggle, then climb upwards, watching as wars, pandemics, economic booms, all play out on the display of life expectancy, income levels, etc. It's not just data. It's real life.

Dataset

I have used the World Bank dataset on life expectancy and GDP per capita from 1960 to 2023. The data was in a messy state, i.e., useless data in some columns, missing values everywhere, years in column names, and aggregated values for regions as well as countries. The 70 year columns were converted into two columns, one named 'Year' and the other the actual value, using the 'melt' function. Then the datasets are merged based on country and year, and only those are taken where both variables are available. After filtering out impossible values, i.e., life expectancy less than 20 or greater than 90, the dataset contains 11,366 observations and 214 countries and regions.

Why This Visualization

I decided to use an animated scatter plot in a classic Gapminder format for a number of different reasons. First, I think that animation helps to visualize the long journey that countries have taken over these last six decades or so, rather than just seeing them in a static state. I didn’t use a bar graph because that would have required more than 60 bars per country, which would have been complete visual chaos. I didn’t use a line graph because that would have required 214 lines, which would have resembled a plate of spaghetti. I didn’t use a scatter plot because that only shows one year, and I didn’t want to represent the entire story. I wanted to be able to feel the progress that these countries have made as they move across the screen.

Design Decisions

There are several key design decisions I made that were very important. The first is that I decided to use a logarithmic scale for my graph because I am working with a range from 50 to over 250,000. There is a 5,000-fold difference in the data. If I were to use a linear scale, all of the developing countries would be squished into a small dot on the left side of the graph.

I decided to use the default colors from Plotly because they are accessible for a colorblind user, are different enough to distinguish between seven regions, and are easy on the eyes. I decided to use a different size for my dots based on their GDP per capita. The bigger the country, the bigger the dot.

Fourth, I added dashed horizontal lines at 50 and 70 years of life expectancy. The line at 50 represents the "surviving childhood" threshold, which many developing nations surpassed during the 1960s and 70s. The line at 70 represents the "developed nation" threshold. These provide the viewer instant guidance.

Fifth, I added annotations noting Japan has achieved 84.8 as of 2023, the struggles of Sub-Saharan Africa, and the impact of COVID-19. Without these, the viewer has no idea what they are seeing as the dots move. Finally, the intervals are set at 5 years, as testing showed that using every year would be too fast, and every decade would ignore the COVID-19 impact.

Key Insights

The data provided some significant findings. First, the world is dramatically better than most people think. Life expectancy in the world has increased from 52.3 years in 1960 to 73.8 years in 2023, an increase of 21.5 years. The world GDP per capita has increased from $2,100 to $12,000, an increase of 5.7 times. In other words, your great grandparents probably died in their early fifties, and you are likely to live into your mid-seventies. That is an extra 22 years with your family.

Second, East Asia experienced a miracle, gaining 28.4 years in life expectancy, which is like having an extra lifetime. China went from famine at 36 years to 78 years now. South Korea went from 53 years to 84 years. India went from 42 years to 70 years, even though its GDP per capita only rose modestly. This proves that you don't have to be rich to be healthy—smart policy helps enormously.

Third, Sub-Saharan Africa rose from 40.5 to 64.6 years, an increase of 24.1 years, more than the world average. However, even after such dramatic improvements, Sub-Saharan Africa still lags behind where Europe was in 1960. Outliers: Botswana went down because of HIV/AIDS and then rebounded. Rwanda went from 41 years because of genocide to 69 years now.

Fourth, COVID-19 was unprecedented. Global life expectancy declined by 1.8 years. World War II was not enough to cause this effect worldwide. Ecuador declined by 5.3 years, Bolivia by 4.9 years, Mexico by 4.1 years, and the US by 2.7 years, making America the worst performer among developed nations. East Asia and Australia were most resilient, proving that lockdowns, masks, and public trust in governments and universal healthcare saved lives.

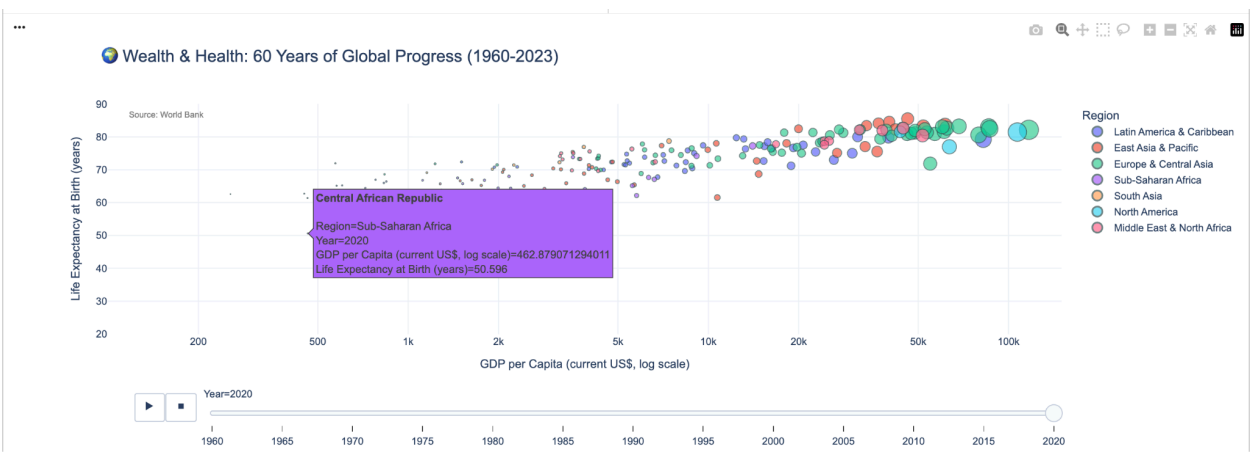
Fifth, money is helpful but not everything. At levels below $2,000, every dollar makes a tremendous difference. At levels above $10,000, every dollar makes little difference. Outliers

exist to validate this. Cuba scores 78 years at $9,500 GDP, the same as the US at $70,000. Costa Rica scores 80 years at $13,000 GDP, beating many other countries that have more. The way money is spent is more important than the amount of money.

Final Reflection

The world is a much different place, and it's changed unevenly. A child born today in Japan will likely live into the 22nd century. A child born today in Chad may never see their 60th birthday. The same world, two completely different worlds. The numbers tell us that if you invest in healthcare and education, people will live longer. If you don't, people will die younger. That's what this is all about, not to make pretty charts, but to tell true stories about real people's lives.

Figure 1: Wealth vs Health: Global Trends in GDP per Capita and Life Expectancy (1960–2023)



GitHubLink:
https://github.com/zulaikha210072/Practical-DataScience-Course_Midterm-Dataviz